

Sayan Bhowmik

October 2026

School of Chemical and Biomolecular Engineering
Georgia Institute of Technology
Ford ES&T Building, 311 Ferst Drive NW
Atlanta, GA 30332, USA
sbhowmik9@gatech.edu

sayanbhowmik.github.io
[LinkedIn](#)
[GitHub](#)
[Google Scholar](#)

EDUCATION

- 2027 Ph.D., School of Chemical and Biomolecular Engineering, Georgia Institute of Technology, Atlanta, GA, USA (expected)
Dissertation Advisors: Phanish Suryanarayana and Andrew J. Medford
- 2022 M.Tech., Department of Chemical Engineering, Indian Institute of Science, Bengaluru, India
- 2020 B.E., Department of Chemical Engineering, Jadavpur University, Kolkata, India

PUBLICATIONS

Refereed Journal Articles

- 2026 P. Ray, **S. Bhowmik**, P. Suryanarayana, S. R. Kalidindi, A. J. Medford, "Electronic manifolds for extrapolative alloy discovery," *Digital Discovery* 5, 3850–3862.
- 2025 **S. Bhowmik**, A. Sharma, A. J. Medford, J. E. Pask, P. Suryanarayana, "Ab initio study of strain-driven vacancy clustering in aluminum," *Phys. Rev. B* 112, 174105.
- S. Bhowmik**, A. J. Medford, P. Suryanarayana, "Real-space Hubbard-corrected density functional theory," *J. Chem. Phys.* 163, 234113.
- S. Bhowmik**, J. E. Pask, A. J. Medford, P. Suryanarayana, "Spectral scheme for atomic structure calculations in density functional theory," *Comput. Phys. Commun.* 308, 109448.
- S. Bhowmik**, A. Govind Rajan, "Size and chemical environment control nanopore geometry in 2D MoS₂: From irregular to triangular defects," *Small*, 2504611.
- 2023 **S. Bhowmik**, J. H. Warner, A. Govind Rajan, "Role of chemical etching in the nucleation of nanopores in 2D MoS₂: Insights from first-principles calculations," *J. Phys. Chem. C* 127, 6873–6883.
- 2022 **S. Bhowmik**, A. Govind Rajan, "Chemical vapor deposition of 2D materials: A review of modeling, simulation, and machine learning studies," *iScience*, 103832.

- 2021 **S. Bhowmik**, V. Chakraborty, P. Das, “Batch adsorption of indigo carmine on activated carbon prepared from sawdust: A comparative study and optimization of operating conditions using response surface methodology,” *Results in Surfaces and Interfaces* 3, 100011.

Manuscripts in Submission

- 2026 **S. Bhowmik**, A. J. Medford, P. Suryanarayana, “Bulk boundary condition for surface calculations in density functional theory,” submitted to *J. Chem. Theory Comput.*, [arXiv:2607.07894](https://arxiv.org/abs/2607.07894).

Web-based Publications

- 2026 A. J. Medford, N.-K. Yu, **S. Bhowmik**, “Replacing exams with ‘write your own problem’ assignments in an online data analytics course,” in D. A. Joyner and N. Roy (Eds.), *Georgia Tech Academic AI Playbook*, 2026 ed., pp. 241–244, Georgia Institute of Technology, hdl.handle.net/1853/81982.
- 2024 **S. Bhowmik**, “Electronic Structure Theory,” 14-lecture open-access video series, [Medford Group Training Materials](#), Georgia Institute of Technology.

AWARDS

- 2026 CRE Travel Award, Catalysis and Reaction Engineering Division, American Institute of Chemical Engineers
- Best Online Teaching Assistant Award, Center for Teaching and Learning, Georgia Institute of Technology
- 2024 Shell Outstanding Teaching Assistant Award, Georgia Institute of Technology
- 2023 Best Ph.D. Qualifying Exam Award, Georgia Institute of Technology
- Exemplary Academic Achievement Award, Georgia Institute of Technology
- 2022 Prof. N. R. Kuloor Memorial Award (Gold Medal), Indian Institute of Science

GRANTS AND FELLOWSHIPS

- 2026 Eckert Graduate Fellowship, Georgia Institute of Technology
- 2020 Graduate Aptitude Test in Engineering (GATE) Fellowship, Ministry of Education, Government of India, Indian Institute of Science

CONFERENCE ACTIVITY

Papers Presented

- 2026 “Beyond vacuum padding: Bulk-boundary conditions for accurate semi-periodic density functional theory,” AIChE Annual Meeting, Minneapolis, MN, USA, November 2026.
- “Real-space Hubbard-corrected density functional theory and Hubbard parameters from hybrid functionals,” ACS Spring Meeting, Atlanta, GA, USA, March 2026.
- “From irregular defects to perfect triangles: Tuning nanopore geometry in 2D MoS₂ via size and chemical environment control,” ACS Spring Meeting, Atlanta, GA, USA, March 2026.
- 2025 “Ab initio investigation into the aggregation of vacancies and formation of dislocation loops in aluminum,” Society of Engineering Science Annual Technical Meeting, Atlanta, GA, USA, October 2025.
- 2024 “Ab initio investigation into the aggregation of vacancies and formation of dislocation loops in aluminum,” Gordon Research Conference on Computational Materials Science and Engineering, Newry, ME, USA, July 2024. Poster.
- “SPARC-atom: An efficient real-space electronic structure calculator for atoms,” Gordon Research Seminar on Computational Materials Science and Engineering, Newry, ME, USA, July 2024. Poster.

TEACHING EXPERIENCE

Georgia Institute of Technology, Lecturer, Density Functional Theory Module

Vertically Integrated Projects: Big Data and Quantum Mechanics (Fall and Spring, 2024–present)

Georgia Institute of Technology, Teaching Assistant

Data Analytics for Chemical Engineers (Fall 2025)

Transport Phenomena (Fall 2023)

RESEARCH EXPERIENCE

- 2022–present Graduate Research Assistant, School of Chemical and Biomolecular Engineering, Georgia Institute of Technology. Advisors: Phanish Suryanarayana and Andrew J. Medford.
- Real-space density functional theory method development in the SPARC electronic structure code, including bulk boundary conditions for surface calculations, Hubbard-corrected density functional theory with a spectral atomic solver, and strain-driven vacancy clustering in aluminum.
- 2021–2022 Graduate Research Fellow, Department of Chemical Engineering, Indian Institute of Science. Advisor: Ananth Govind Rajan.

Nanopore nucleation and geometry in 2D MoS₂ using density functional theory, Marcus theory, kinetic Monte Carlo, and chemical graph theory.

SERVICE TO PROFESSION

Workshop Instructor, SPARC hands-on training session, 10th i-CoMSE Workshop on QM/MM and Ab Initio Molecular Dynamics, Oklahoma State University, Stillwater, OK, USA, 2025

DEPARTMENTAL AND UNIVERSITY SERVICE

Reviewer, Presidential Undergraduate Research Award, Georgia Institute of Technology, 2024–2025

REFERENCES

Phanish Suryanarayana
Professor, School of Civil and Environmental Engineering and School of Computational Science and Engineering
Georgia Institute of Technology, Atlanta, GA 30332, USA
phanish.suryanarayana@ce.gatech.edu

Andrew J. Medford
Associate Professor and J. Frank Scarborough Faculty Fellow, School of Chemical and Biomolecular Engineering
Georgia Institute of Technology, Atlanta, GA 30332, USA
ajm@gatech.edu

Ananth Govind Rajan
Associate Professor, Department of Chemical Engineering
Indian Institute of Science, Bengaluru 560012, India
ananthgr@iisc.ac.in